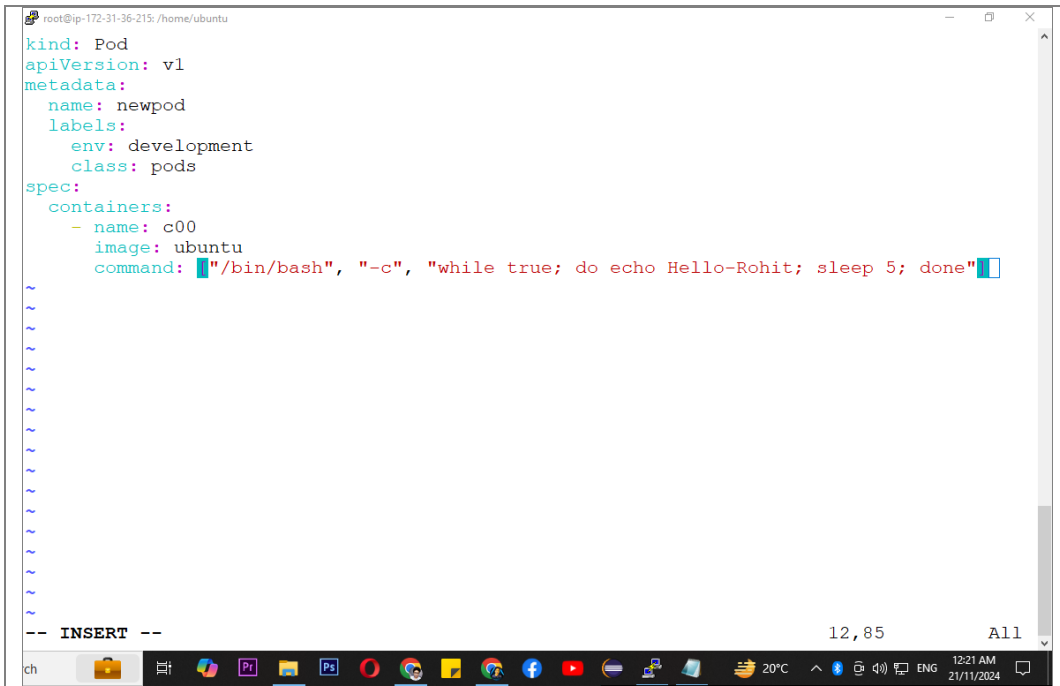

Labels and Selectors in Kubernetes

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Manifest with labels



```
kind: Pod
apiVersion: v1
metadata:
  name: newpod
  labels:
    env: development
    class: pods
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
```

1.

(This manifest defines a Kubernetes Pod named newpod that runs a single container based on the ubuntu image. The container executes a command that continuously prints "Hello-Rohit" every 5 seconds. Labels are key-value pairs that can be used to organize and select resources. In this case, the Pod has two labels: env: development and class: pods.)

```
kind: Pod
apiVersion: v1
metadata:
  name: newpod
  labels:
    env: development
    class: pods
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
```

How to check the lables

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
leo           1/1     Running   0           36s
testpod3      3/3     Running   0           9s
root@ip-172-31-36-215:/home/ubuntu# vi labels.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
leo           1/1     Running   0          112s   <none>
testpod3      3/3     Running   0          85s    <none>
root@ip-172-31-36-215:/home/ubuntu# A kubectl apply -f labels.yml
pod/newpod created
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
leo           1/1     Running   0          2m27s   <none>
newpod        1/1     Running   0           4s     B class=pods,env=development
testpod3      3/3     Running   0           2m      <none>
root@ip-172-31-36-215:/home/ubuntu#
```

2.

`kubectl get pods --show-labels`

How to give labels using Imperative method

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
leo           1/1     Running   0           36s
testpod3      3/3     Running   0           9s
root@ip-172-31-36-215:/home/ubuntu# vi labels.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
leo           1/1     Running   0          112s   <none>
testpod3      3/3     Running   0          85s    <none>
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f labels.yml
pod/newpod created
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
leo           1/1     Running   0          2m27s   <none>
newpod        1/1     Running   0           4s     A class=pods,env=development
testpod3      3/3     Running   0           2m      <none>
root@ip-172-31-36-215:/home/ubuntu# kubectl label pods newpod myname=rohit
pod/newpod labeled
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
leo           1/1     Running   0          5m17s   <none>
newpod        1/1     Running   0          2m54s   B class=pods,env=development,myname=rohit
testpod3      3/3     Running   0          4m50s   <none>
root@ip-172-31-36-215:/home/ubuntu#
```

3.

`kubectl label pods pod_name key=value (for ex: myname=rohit)`

Equality based Label Selectors(=, !=)

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l env=development
NAME      READY   STATUS    RESTARTS   AGE
newpod    1/1     Running   0           5m55s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l env!=development
NAME      READY   STATUS    RESTARTS   AGE
leo       1/1     Running   0           8m31s
testpod3  3/3     Running   0           8m4s
root@ip-172-31-36-215:/home/ubuntu#
```

4.

`kubectl get pods -l key=value`
`kubectl get pods -l key!=value`

How to delete pods using lables & selectors

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l env=development
NAME      READY   STATUS    RESTARTS   AGE
newpod    1/1     Running   0           5m55s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l env!=development
NAME      READY   STATUS    RESTARTS   AGE
leo       1/1     Running   0           8m31s
testpod3  3/3     Running   0           8m4s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
leo       1/1     Running   0           10m
newpod    1/1     Running   0           7m48s
testpod3  3/3     Running   0           9m44s
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pod -l env=development
pod "newpod" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
leo       1/1     Running   0           11m
testpod3  3/3     Running   0           10m
root@ip-172-31-36-215:/home/ubuntu#
```

5.

`kubectl delete pod -l key=value`

Set based Label Selectors(in, notin, exists, doesnotexist)

```
root@ip-172-31-36-215: /home/ubuntu
newpod 1/1 Running 0 5m55s
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods -l env!=development
NAME READY STATUS RESTARTS AGE
leo 1/1 Running 0 8m31s
testpod3 3/3 Running 0 8m4s
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods
NAME READY STATUS RESTARTS AGE
leo 1/1 Running 0 10m
newpod 1/1 Running 0 7m48s
testpod3 3/3 Running 0 9m44s
root@ip-172-31-36-215: /home/ubuntu# kubectl delete pod -l env=development
pod "newpod" deleted
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods
NAME READY STATUS RESTARTS AGE
leo 1/1 Running 0 11m
testpod3 3/3 Running 0 10m
root@ip-172-31-36-215: /home/ubuntu# kubectl apply -f labels.yml
pod/newpod created
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods -l 'env in(development,testing)'
NAME READY STATUS RESTARTS AGE
newpod 1/1 Running 0 77s
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods -l 'env not in(development,testing)'
Error from server (BadRequest): Unable to find "/v1, Resource=pods" that match label selector in(development,testing)", field selector "": unable to parse requirement: found 'not', , notin, =, ==, !=, gt, lt
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods -l 'env notin(development,testing)'
NAME READY STATUS RESTARTS AGE
leo 1/1 Running 0 15m
testpod3 3/3 Running 0 14m
root@ip-172-31-36-215: /home/ubuntu#
```

6.

```
kubectl get pods -l 'key in(value1,value2)'
kubectl get pods -l 'key notin(value1,value2)'
```

How to search for pods using labels and selectors

```
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods -l class=pods
NAME READY STATUS RESTARTS AGE
newpod 1/1 Running 0 6m7s
root@ip-172-31-36-215: /home/ubuntu#
```

7.

```
kubectl get pods -l class=pods
```

How to delete pods using set based label selectors



A terminal window showing the process of deleting a pod using a set-based label selector. The user first runs `kubectl get pods -l class=pods`, which displays a table with columns NAME, READY, STATUS, RESTARTS, and AGE. The table shows one pod named 'newpod' in a 'Running' state. Then, the user runs `kubectl delete pods -l 'env in(development)'`, and the terminal confirms 'pod "newpod" deleted'.

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l class=pods
NAME      READY   STATUS    RESTARTS   AGE
newpod    1/1     Running   0           1m7s
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pods -l 'env in(development)'
pod "newpod" deleted
root@ip-172-31-36-215:/home/ubuntu#
```

8.

```
kubectl delete pods -l 'key in(value)'
```

Manifest with a nodeSelector

```
root@ip-172-31-36-215: /home/ubuntu
kind: Pod
apiVersion: v1
metadata:
  name: nodelabels
  labels:
    env: development
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
  nodeSelector:
    hardware: t2-medium

-- INSERT --
```

9.

(This manifest defines a Kubernetes Pod named nodelabels that runs a single container based on the ubuntu image.

The container executes a command that continuously prints "Hello-Rohit" every 5 seconds.

The Pod is scheduled on nodes labeled with hardware: t2-medium. This specifies the node selection criteria for the Pod.

The Pod will be scheduled on nodes that have the label hardware: t2-medium.

Make sure that you have nodes in your cluster with this label.)

```
kind: Pod
apiVersion: v1
metadata:
  name: nodelabels
  labels:
    env: development
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
  nodeSelector:
    hardware: t2-medium
```

Error creating the pod (nodeSelector)

```
root@ip-172-31-36-215: /home/ubuntu
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -l class=pods
NAME      READY   STATUS    RESTARTS   AGE
newpod    1/1     Running   0           6m7s
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pods -l 'env in(development)'
pod "newpod" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube   Ready    control-plane   3h    v1.31.0
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f newmanifest.yml
pod/nodelabels created
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
nodelabels 0/1     Pending   0           7s
testpod3   3/3     Running   0           41m
root@ip-172-31-36-215:/home/ubuntu#
```

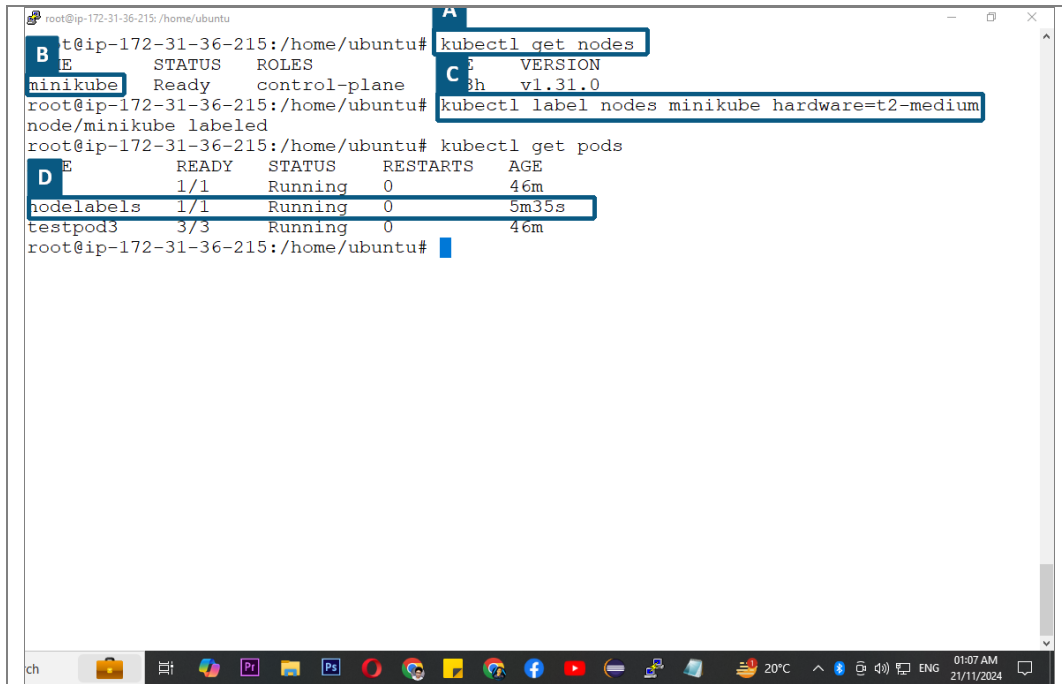
10.

Node cannot be scheduled cause we haven't added the nodeSelector hardware label to the node

```
Port:      <none>
Host Port: <none>
Command:
  /bin/bash
  -c
  while true; do echo Hello-Rohit; sleep 5; done
Environment: <none>
Mounts:
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-zvbtd (ro)
Conditions:
  Type           Status
  PodScheduled   False
Volumes:
  kube-api-access-zvbtd:
    Type:              Projected (a volume that contains injected data from multip
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    ConfigMapOptional:  <nil>
    DownwardAPI:        true
    Class:              BestEffort
  node-Selectors:      hardware=t2-medium
Tolerations:
  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type      Reason              Age   From                  Message
  --      -
  Warning   FailedScheduling    97s   default-scheduler    0/1 nodes are available: 1 node(s)
  Pod's node affinity/selector.
  precondition: 0/1 nodes are available: 1 Preemption is not
  scheduling.
```

11.

How to add the nodeSelector hardware label to the node



```

root@ip-172-31-36-215:/home/ubuntu# kubectl get nodes
NAME              STATUS    ROLES    AGE   VERSION
minikube          Ready     control-plane  3h   v1.31.0
root@ip-172-31-36-215:/home/ubuntu# kubectl label nodes minikube hardware=t2-medium
node/minikube labeled
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME              READY   STATUS    RESTARTS   AGE
nodeLabels        1/1     Running   0           46m
testpod3          3/3     Running   0           46m
root@ip-172-31-36-215:/home/ubuntu#

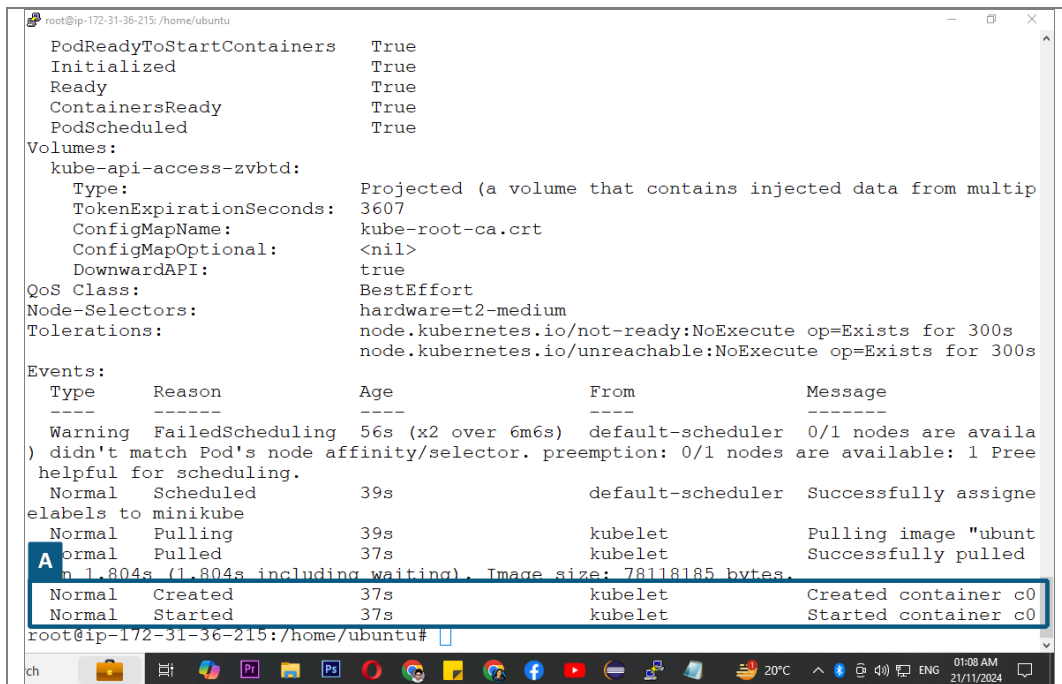
```

12.

```
kubectl get nodes
```

```
kubectl label nodes node_name hardware=t2-medium(key-value)
```

Node is successfully scheduled and container is created



```

PodReadyToStartContainers    True
Initialized                  True
Ready                        True
ContainersReady              True
PodScheduled                  True
Volumes:
  kube-api-access-zvbtd:
    Type:                Projected (a volume that contains injected data from multip
    TokenExpirationSeconds: 3607
    ConfigMapName:        kube-root-ca.crt
    ConfigMapOptional:    <nil>
    DownwardAPI:          true
QoS Class:                   BestEffort
Node-Selectors:               hardware=t2-medium
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason             Age   From          Message
  ----     -
  Warning   FailedScheduling   56s   (x2 over 6m6s) default-scheduler 0/1 nodes are availa
) didn't match Pod's node affinity/selector. preemption: 0/1 nodes are available: 1 Pre
helpful for scheduling.
  Normal    Scheduled          39s   default-scheduler Successfully assigne
elabels to minikube
  Normal    Pulling            39s   kubelet        Pulling image "ubunt
  Normal    Pulled             37s   kubelet        Successfully pulled
n 1.804s (1.804s including waiting). Image size: 78118185 bytes.
  Normal    Created            37s   kubelet        Created container c0
  Normal    Started            37s   kubelet        Started container c0
root@ip-172-31-36-215:/home/ubuntu#

```

13.